Huawei Access Node Optical Transport Failure Resolution SOP (KR-ACC-41)

Node: KR-ACC-41 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_7665

Description

This Standard Operating Procedure (SOP) outlines the technical steps to resolve Optical Transport Failure on Huawei Access Node KR-ACC-41 at KR Puram Access Site. The failure domain in question is attributed to SFP/XFP degradation, fiber cuts, or Rx/Tx power loss due to the inherent design of the Huawei Access Node's Optical Transport Unit (OTU). The OTU, responsible for transmitting optical signals between the node and the Optical Transport Network (OTN), utilizes SFP/XFP transceivers to convert electrical signals into optical signals and vice versa. In the event of SFP/XFP degradation, fiber cuts, or Rx/Tx power loss, the OTU's ability to transmit optical signals is compromised, resulting in the Optical Transport Failure. The OTU's functionality is further complicated by the OTN protocol, which requires accurate timing and synchronization between nodes to ensure seamless transmission of optical signals. The loss of timing and synchronization due to SFP/XFP degradation or fiber cuts can cause the OTU to malfunction, leading to the Optical Transport Failure. Therefore, it is essential to diagnose and resolve the root cause of the failure to prevent further network disruption.

Pre-requisite Checks

- Verify that the issue is not related to any other components or services within the network.

- Ensure that the power supply to the node is stable and within the recommended operating range.

- Check the environmental conditions at the site, including temperature and humidity levels, to ensure they are within the recommended operating range.

- Verify that the redundant link is available and functioning correctly.

- Dispatch a remote hand to the KR Puram Access site to assist with the troubleshooting process.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display optical transport unit, Expected Output: Optical Transport Unit information will be displayed, including the current status of the OTU, SFP/XFP information, and received/transmitted optical signal levels., Description: This command will provide information on the current status of the OTU, including the current
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status of the OTU, SFP/XFP information, and received/transmitted
optical signal levels. This will help identify if the issue is related
to the OTU or the SFP/XFP transceiver.

Command: display optical signal trace, Expected Output: Optical signal
trace information will be displayed, including the current status of
the optical signals and any errors detected., Description: This
command will provide information on the current status of the optical
signals, including any errors detected. This will help identify if the
issue is related to the fiber or the OTU.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP transceiver, Command: replace optical transport unit transceiver, Description: Replace the SFP/XFP transceiver with a new one, ensuring that the correct type and version are used. Verify that the SFP/XFP transceiver is securely seated in the OTU and the optical signal levels are within the recommended range.

Step 2: Step: Verify fiber connections, Command: verify fiber connections, Description: Verify that the fiber connections are secure and not damaged. Check the fiber connections for any signs of wear or damage, and replace them if necessary.

Step 3: Step: Adjust Rx/Tx power levels, Command: adjust optical signal power, Description: Adjust the Rx/Tx power levels to ensure they are within the recommended range. Verify that the optical signal levels are stable and within the recommended range.

Post-Check Verification

```
Command: display optical transport unit, Expected Output: Optical
Transport Unit information will be displayed, including the current
status of the OTU, SFP/XFP information, and received/transmitted
optical signal levels., Description: Verify that the OTU is
functioning correctly and the SFP/XFP transceiver is securely seated.

Command: display optical signal trace, Expected Output: Optical signal
trace information will be displayed, including the current status of
the optical signals and any errors detected., Description: Verify that
the optical signals are stable and within the recommended range.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP transceiver, Command: restore original optical transport unit transceiver, Description: Restore the original SFP/XFP transceiver and verify that the OTU is functioning correctly.

- Step: Verify fiber connections, Command: verify fiber connections, Description: Verify that the fiber connections are secure and not damaged.